
Public Spending In New York Towns

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Question:

Does government spending within NY towns come more from local taxes or government aid? (as seen through 3 metrics)

Specifically measuring:

- General Government Spending
- Public Safety
- Social Services

Data

Data Source: Office of the New York State Comptroller, Local Government Bulk Data

- 9981 observations across 914 NY Towns
- 11 years worth of data (2013–2023)

Converted to panel by appending each year into one dataset

gen_gov

Total amount of expenditures for services provided by the governmental entity for the benefit of the public or governmental body as a whole.

Includes:

- Administration
- Zoning and planning
- Operations
- Judgments
- County distribution of sales tax
- Miscellaneous general government

public_safety

Total amount of expenditures for police, fire and other public safety services.

Includes:

- Public safety administration
- Police
- Fire protection
- Emergency response
- Correctional services
- Disaster response
- Homeland security and civil defense
- Miscellaneous public safety

social_services

Total amount of expenditures for providing public assistance for individuals/families economically unable to provide essential needs for themselves.

Includes:

- Social service administration
- Financial assistance
- Medicaid
- Non-Medicaid medical assistance
- Housing assistance, employment services
- Youth services
- Public facilities
- Miscellaneous social services

fed_aid

Total amount of revenues derived from the federal government

Includes:

- Assistance with education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous federal Aid

state_aid

Total amount of revenues derived from State Aid.

Includes:

- Education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous State Aid

local_taxes

Total amount of tax revenue within town.

Includes:

- Property Tax
- Non-Property Tax (city income tax)
- Other property Tax (gains from sale of tax-acquired property)
- Sales & Use Tax (Sales Tax)

Clustering Standard Errors

Looks at each individual entity's standard error instead of lumping them all together.

- Not clustering underestimates standard errors

Coefficients don't change when clustering, only standard errors

Clustering Standard Errors

With Clustered SE:

gen_gov	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
fed_aid	20.54548	3.27292	6.28	0.000	14.12142	26.96955
state_aid	23.39038	8.584306	2.72	0.007	6.541171	40.23959
local_taxes	-.956704	.6695676	-1.43	0.153	-2.270926	.357518
calendar_year						
2014	2499028	2363468	1.06	0.291	-2139968	7138024
2015	1848775	2340137	0.79	0.430	-2744428	6441978
2016	2712811	2514260	1.08	0.281	-2222158	7647780
2017	-3697988	3428213	-1.08	0.281	-1.04e+07	3030881
2018	-4038495	5619114	-0.72	0.473	-1.51e+07	6990658
2019	-6561880	5381499	-1.22	0.223	-1.71e+07	4000885
2020	-7254732	7048451	-1.03	0.304	-2.11e+07	6579912
2021	4151582	6145287	0.68	0.499	-7910338	1.62e+07
2022	-2700332	5738988	-0.47	0.638	-1.40e+07	8564108
2023	2601491	7255519	0.36	0.720	-1.16e+07	1.68e+07
_cons	-1.87e+07	5542937	-3.38	0.001	-2.96e+07	-7855774
sigma_u	52823354					
sigma_e	65090374					
rho	.39708005	(fraction of variance due to u_i)				

Without Clustered SE:

gen_gov	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
fed_aid	20.54548	.6007752	34.20	0.000	19.36731	21.72366
state_aid	23.39038	1.409932	16.59	0.000	20.62538	26.15538
local_taxes	-.956704	.1379403	-6.94	0.000	-1.227217	-.6861907
calendar_year						
2014	2499028	6260933	0.40	0.690	-9779218	1.48e+07
2015	1848775	6324292	0.29	0.770	-1.06e+07	1.43e+07
2016	2712811	6574661	0.41	0.680	-1.02e+07	1.56e+07
2017	-3697988	6952223	-0.53	0.595	-1.73e+07	9935939
2018	-4038495	6625908	-0.61	0.542	-1.70e+07	8955500
2019	-6561880	6987025	-0.94	0.348	-2.03e+07	7140297
2020	-7254732	6814128	-1.06	0.287	-2.06e+07	6108379
2021	4151582	6258424	0.66	0.507	-8121744	1.64e+07
2022	-2700332	6061453	-0.45	0.656	-1.46e+07	9186717
2023	2601491	6217040	0.42	0.676	-9590677	1.48e+07
_cons	-1.87e+07	4673745	-4.01	0.000	-2.79e+07	-9569777
sigma_u	52823354					
sigma_e	65090374					
rho	.39708005	(fraction of variance due to u_i)				

* Note regression coefficients do not change when clustering standard errors, only standard errors (red) and corresponding T statistics (blue)

Clustering Standard Errors

	(1) gen_gov	(2) gen_gov	(3) gen_gov
fed_aid	17.92*** (36.23)	20.55*** (34.20)	20.55*** (6.28)
state_aid	26.77*** (22.06)	23.39*** (16.59)	23.39** (2.72)
local_taxes	-1.468*** (-12.48)	-0.957*** (-6.94)	-0.957 (-1.43)
2013.calen~r		0 (.)	0 (.)
2014.calen~r		2499028.0 (0.40)	2499028.0 (1.06)
2015.calen~r		1848774.9 (0.29)	1848774.9 (0.79)
2016.calen~r		2712810.6 (0.41)	2712810.6 (1.08)
2017.calen~r		-3697988.1 (-0.53)	-3697988.1 (-1.08)
2018.calen~r		-4038494.8 (-0.61)	-4038494.8 (-0.72)

(1) Random Effects

(2) FE (no cluster)

(3) FE (cluster)

2017.calen~r	-3697988.1 (-0.53)	-3697988.1 (-1.08)	
2018.calen~r	-4038494.8 (-0.61)	-4038494.8 (-0.72)	
2019.calen~r	-6561879.7 (-0.94)	-6561879.7 (-1.22)	
2020.calen~r	-7254731.9 (-1.06)	-7254731.9 (-1.03)	
2021.calen~r	4151582.2 (0.66)	4151582.2 (0.68)	
2022.calen~r	-2700332.4 (-0.45)	-2700332.4 (-0.47)	
2023.calen~r	2601490.8 (0.42)	2601490.8 (0.36)	
_cons	-14813148.0*** (-8.45)	-18735405.9*** (-4.01)	-18735405.9*** (-3.38)
N	2964	2964	2964
t statistics in parentheses			
* p<0.05, ** p<0.01, *** p<0.001			

* Note the change in statistical significance for local_taxes before clustering standard errors and after (highly significant to insignificant after cluster)

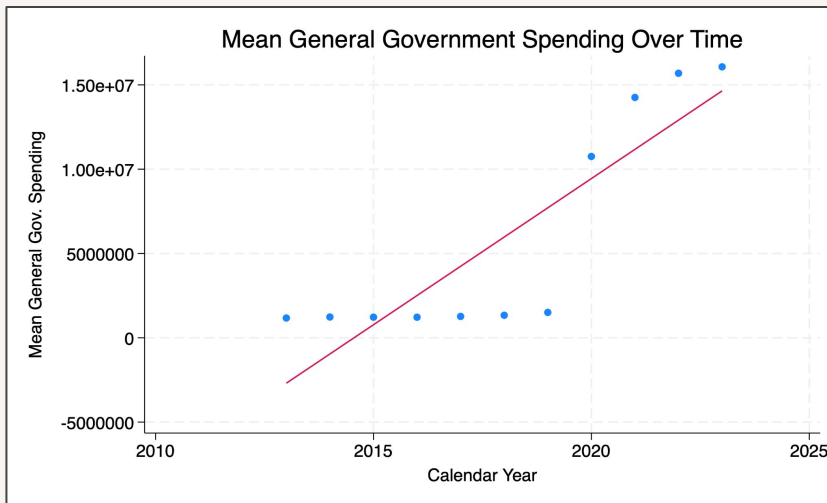
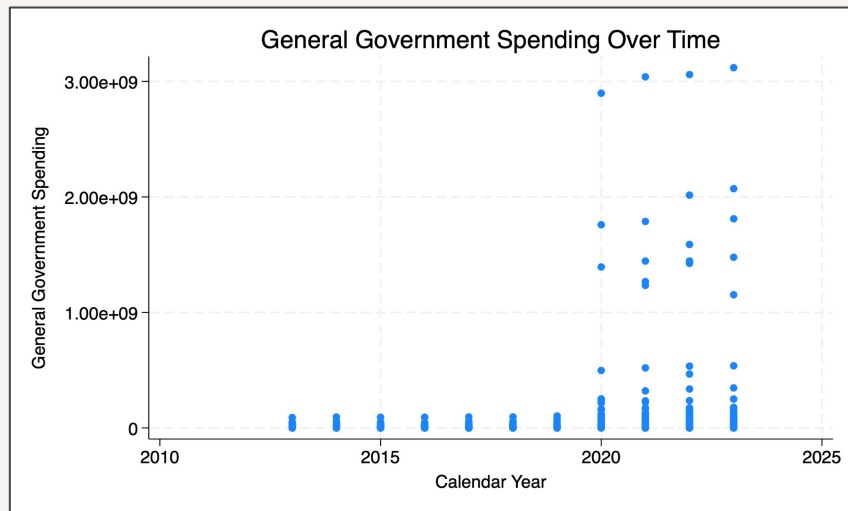
Clustering Standard Errors

Potential Unit Fixed Effects:

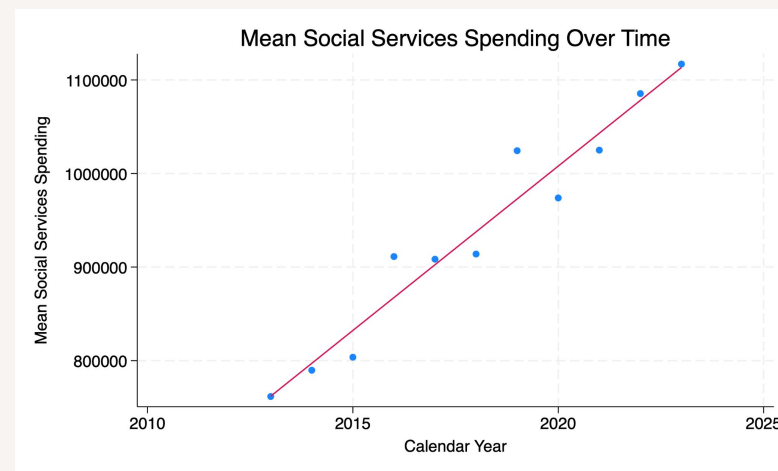
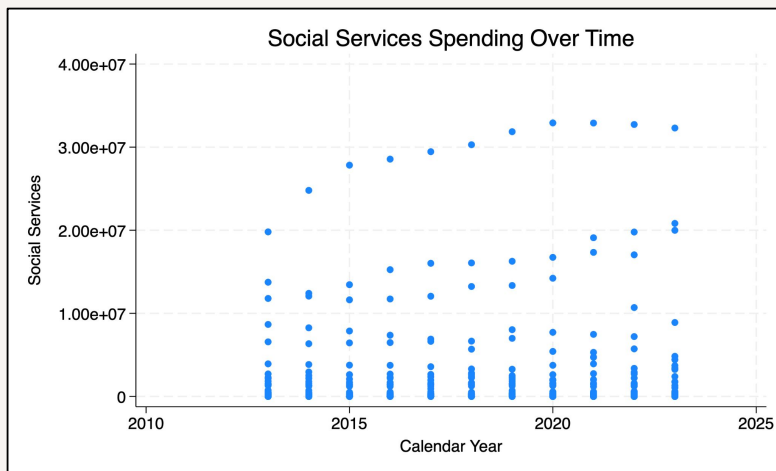
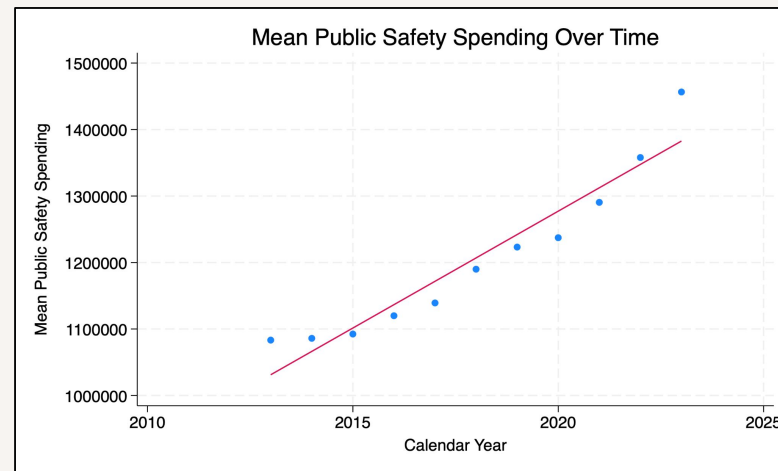
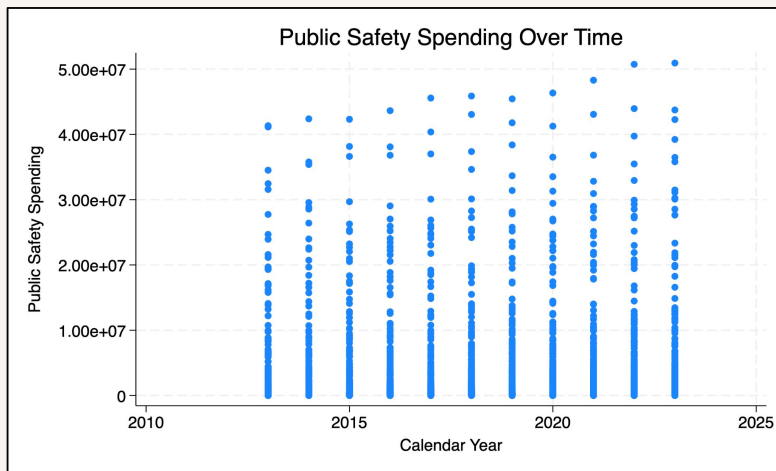
- Sense of community within town
- Urban vs Rural status
- Vacation town status

Potential Time Fixed Effects:

- COVID
- Political Regime changes
- Spending Contributions to Pensions (Large spike in 2019)



* *Graphs are shown in dollars, not thousands*



* Graphs are shown in dollars, not thousands

Findings

	(1) gen_gov	(2) public_saf~y	(3) social_ser~s
fed_aid	20.55*** (6.28)	-0.0759 (-1.73)	0.189** (2.97)
state_aid	23.39** (2.72)	-0.878*** (-5.60)	-0.199** (-2.72)
local_taxes	-0.957 (-1.43)	0.275*** (12.40)	0.0213*** (3.74)
2013.calen~r	0 (.)	0 (.)	0 (.)
2014.calen~r	2499028.0 (1.06)	-122505.3 (-1.43)	119726.0 (0.94)
2015.calen~r	1848774.9 (0.79)	-135308.6 (-1.17)	203952.6 (1.07)
2016.calen~r	2712810.6 (1.08)	-29084.7 (-0.27)	251482.5 (1.20)
2017.calen~r	-3697988.1 (-1.08)	50542.5 (0.29)	299150.9 (1.15)
2018.calen~r	-4038494.8 (-0.72)	8741.2 (0.04)	719509.2 (1.77)
2019.calen~r	-6561879.7 (-1.22)	277918.1 (1.27)	830983.0 (1.72)

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2017.calen~r	-3697988.1 (-1.08)	50542.5 (0.29)	299150.9 (1.15)
2018.calen~r	-4038494.8 (-0.72)	8741.2 (0.04)	719509.2 (1.77)
2019.calen~r	-6561879.7 (-1.22)	277918.1 (1.27)	830983.0 (1.72)
2020.calen~r	-7254731.9 (-1.03)	242602.1 (1.31)	796100.3 (1.40)
2021.calen~r	4151582.2 (0.68)	635709.4** (2.93)	767891.2 (1.86)
2022.calen~r	-2700332.4 (-0.47)	445771.3* (2.20)	1055117.9 (1.85)
2023.calen~r	2601490.8 (0.36)	457935.5* (2.33)	810872.6 (1.78)
_cons	-18735405.9*** (-3.38)	337351.6* (2.38)	10466.0 (0.03)
N	2964	2964	604
t statistics in parentheses * p<0.05, ** p<0.01, *** p<0.001			

* 2013 is reference year for LSDV Time Fixed Effects

Findings

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General Government spending:
mainly driven by state and federal
aid

Public Safety spending: mainly
driven by local taxes

Social Services spending: mainly
driven by federal aid, but local
taxes do contribute

Findings

Within towns in NY:

- Most of the general spending comes from government aid
- Local taxes are the main contributor to public safety spending
- Federal aid and local taxes contribute highest to social services spending

Extensions:

- Expand the data outside of just NY, possibly Tri-state area or entire US
- Expand data for more years than just from 2013–2023
- Control for more variables (population, political regime, median income)

Thank you